



CS1002 - Programming Fundamentals



Assignment 2

Total Marks: 100

Instructions:

1. Assignment should be submitted in a zip file named as 2XP_XXXX_Name.
2. Create a PDF file with screenshots of code and output.
3. The zip file should contain PDF and all .c files.
4. Code must contain proper indentation and comments else marks will be deducted.
5. **Plagiarism of any shape or form will not be tolerated. In case of plagiarism, the particular question will be marked zero and 50% marks from total obtained marks will be deducted.**
6. In case of an invalid input your function shall return -1.

0. Warm up Question

20 marks

It's party time.

Your parents are out of the city for some important work. Now you are home alone and it's boring. So you decide to throw a party. But unfortunately, you don't have many friends so you decide to call your 'only' friend who has a large friend circle. You ask him to bring his friends to your house for the party but the number should not be more than 15. You also tell him that you will only pay 50% of the total food bill and the rest 50% will be paid by the guests. You order 3 Large pizzas (8 pieces per pizza) and 3 1000 ml (5 glasses per drink) drinks. The total amount is 5000 pkr.

- a) Write two functions “*div_pizza*” and “*div_drink_glasses*” to evenly distribute these food items among the n number of guests and you. The functions shall return the calculated values.

Input	Output
n = 5, 0, -1	
div_pizza(5)	4
div_pizza(0)	24
div_pizza(-1)	-1
div_drink_glasses(5)	2
div_drink_glasses(0)	15
div_drink_glasses(-1)	-1

- b) Now write a function “*div_amount*” that returns the amount each guest has to pay each guest has to pay.

Input	Output
n = 5	
div_amount(n)	500.0

1. Ice cream Price Calculator (Console application)

30 marks

Write a program that calculates the price of an Ice-cream/s purchased by the customer. The program will be deployed in an automated ice-cream store where everything is controlled by machines.

Your program should ask the customer to enter their budget and then suggest ice-creams according to it. It should allow the customer to select from the ice-cream type first. And then from flavor and size.

The program should prompt the user about his/her choice. The user will either continuously enter their choices until there is not enough budget for further purchasing or they can exit the menu explicitly. Finally the program should display the total price of the order with the total number of items purchased.

Ice-cream Types
Cone
Cups

Cone

Flavor	Price(Pkr)
Chocolate	200
Vanilla	250
Strawberry	200
Oreo	250

Cups

Flavor	Small	Medium	Large
Chocolate	200 pkr	250 pkr	300 pkr
Vanilla	250 pkr	300 pkr	350 pkr
Strawberry	200 pkr	250 pkr	300 pkr
Oreo	250 pkr	300 pkr	400 pkr

2. The Game

50 marks

There are two decks of cards, Deck A and Deck B, with numbers written on them. The numbers in each deck range from 0 to 10.

The player is supposed to select one card each from both decks, and beforehand guess the sum of the numbers. Let the guess be a number n . If the guess turns out to be right, the player earns n pkr. Otherwise, the player will lose 10 pkr. The machine plays in the same way, but the machine's guess of the sum is a uniformly distributed number between 3 and 20. The player determines the number of rounds, r , to play, and receives r pkr as initial capital. The winner is the one that has the largest amount of money after r rounds, or the one that avoids to lose all the money.

Functions:

<code>compute_sum();</code>	<code>// return sum of two randomly generated numbers</code>
<code>get_rounds();</code>	<code>// returns the number of rounds</code>
<code>computer_guess();</code>	<code>// return computer's guess</code>
<code>player_guess();</code>	<code>// returns player's guess</code>
<code>player_capital(rounds);</code>	<code>// returns the amount of money left with the player</code>
<code>computer_capital(rounds);</code>	<code>// returns the amount of money left with the computer</code>
<code>play(rounds);</code>	<code>// returns 1 if the player wins else returns 0</code>

PS: Please find the hint on the last page.

Hint: Don't look for hints.